

BASIC MEASUREMENTS OF SHAPES

PROMPT:

Write a Python program to calculate the **radius of a circle, area of a square, and area of a rectangle** using a **while loop and conditional statements (if/elif/else)**. The program should allow the user to select any one of these operations. Use **integer input and display only integer output**.

PROGRAM:

```
while True:
    print("1. Radius of Circle")
    print("2. Area of Square")
    print("3. Area of Rectangle")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        area = int(input("Enter area of circle: "))
        radius = int((area / 3.14) ** 0.5)
        print("Radius =", radius)

    elif choice == 2:
        side = int(input("Enter side of square: "))
        area = side * side
        print("Area of Square =", area)

    elif choice == 3:
        length = int(input("Enter length: "))
        breadth = int(input("Enter breadth: "))
        area = length * breadth
        print("Area of Rectangle =", area)

    elif choice == 4:
        break

    else:
        print("Invalid choice")
```

SAMPLE OUTPUT:

```
1. Area of Circle
2. Area of Square
3. Area of Rectangle
4. Exit
Enter your choice: 1
Enter radius: 25
Area of Circle = 1962

1. Area of Circle
2. Area of Square
3. Area of Rectangle
4. Exit
Enter your choice: 2
Enter side: 4
Area of Square = 16

1. Area of Circle
2. Area of Square
3. Area of Rectangle
4. Exit
Enter your choice: 3
Enter length: 15
Enter breadth: 11
Area of Rectangle = 165

1. Area of Circle
2. Area of Square
3. Area of Rectangle
4. Exit
Enter your choice: 4
Program ended
```